🏦 Chart of Accounts (COA) Engine — Deep HLD Design

> ***Document Focus:*** *COA Engine Architecture · Dynamic Module Integration · Balance Sheet & Income Statement Deep Dive*
> ***Project:*** *GTR Service ERP — Multi-Tenant SaaS Platform*
> ***Reference:*** *Accounting Module (3).docx · GTR Service HLD Architecture*
> ***Style:*** *HLD Design Only — No Code*

Table of Contents

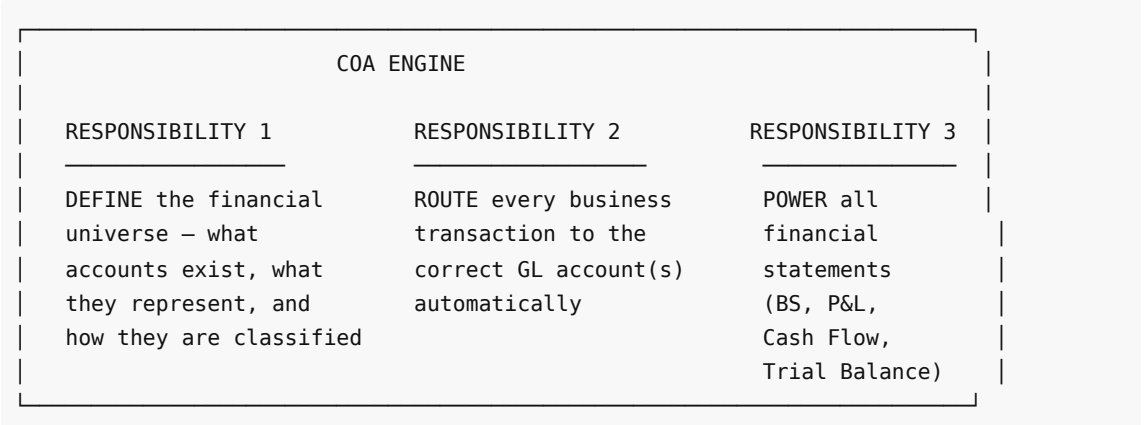
1. What is the COA Engine?
2. COA Engine Anatomy
3. The Three-Bridge System — How COA Connects to Every Module
4. Dynamic GL Resolution Flow
5. Dimension System — Analytical Layer
6. Balance Sheet (BS) — Deep Dive
7. Income Statement (P&L) — Deep Dive
8. How Each Module Feeds the BS & P&L
9. Upcoming Module Integration Design
10. Financial Statement Assembly Engine
11. COA Integration Master Map

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1. What is the COA Engine?

The **Chart of Accounts Engine** is not just a list of accounts. It is the **central financial nervous system** of the entire ERP. Every transaction that happens in every module — a purchase, a sale, a warehouse transfer, a payroll run, a depreciation entry — must eventually resolve to one or more GL accounts in the COA. The COA Engine is the mechanism that makes this resolution possible, automatic, and consistent.

The Three Core Responsibilities of the COA Engine



Why It Is "Dynamic"

The COA Engine is dynamic because the accounts it uses are **not hardcoded**. Instead, they are configured per-tenant, per-company, and per-item through three mapping tables (`ItemGLAccount` , `CompanyGLAccount` , and dimension configurations). This means:

- A **Goods account** for Tenant A can be GL code `2010400` mapped to `BS Inventories`
- The same concept for Tenant B can be GL code `1001100` mapped to `BS Inventories`
- Both post to the same **section** in the Balance Sheet, but use completely different account codes

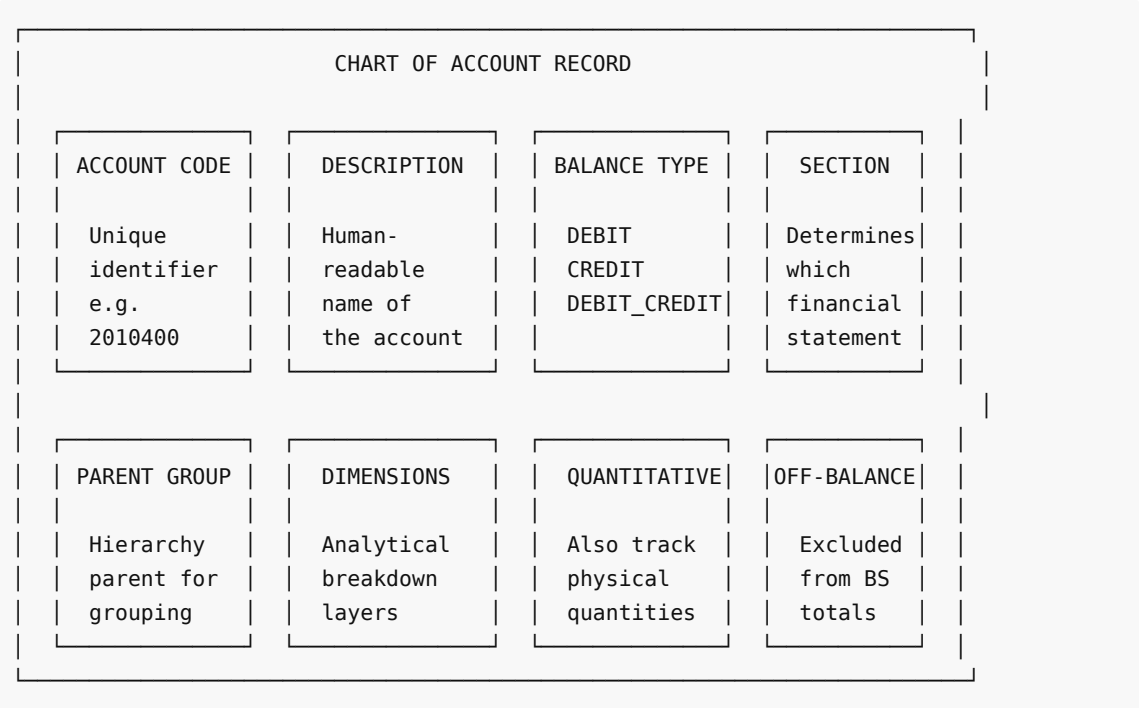
This is true multi-tenant accounting isolation at the GL level.

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2. COA Engine Anatomy

2.1 The Single Account Record

Every account in the COA has five essential attributes that together control its entire behavior in the system:



2.2 Balance Type — The Fundamental Rule

Every account has a **normal balance side**. This drives how the system interprets increases and decreases:

Balance Type	Meaning	Examples
DEBIT	Normal balance is on the Debit side. Debit increases it, Credit decreases it	Assets, Expenses, Cost of Sales
CREDIT	Normal balance is on the Credit side. Credit increases it, Debit decreases it	Liabilities, Equity, Income

DEBIT_CREDIT	Can be either — used for accounts that can swing both ways	Some contra accounts, clearing accounts

> ***Rule enforced by the GL Engine:*** *Every journal entry must have* `Total Debits = Total Credits`*. This is validated by the* `isBalanced()` *check before any entry is posted.*

2.3 Section — The Classification Key

The `section` field on each account is the **most important field** for financial reporting. It is the routing tag that tells the financial statement engine where to place this account's balance.

```
Account Section Value                → Where It Appears
──────────────────────────────────────────────────────────
"BS Inventories"                     → Balance Sheet → Assets → Current Assets
"BS Fixed Assets"                    → Balance Sheet → Assets → Non-Current Assets
"BS Current Trade Payables"          → Balance Sheet → Liabilities → Current
"BS Deferred Revenue"                → Balance Sheet → Liabilities
"BS Cash"                            → Balance Sheet → Assets → Current
"BS Input VAT"                       → Balance Sheet → Assets → Current
"P&L Sales of Goods"                 → Income Statement → Revenue
"P&L COS Goods for Resale"           → Income Statement → Cost of Sales
"P&L Admin Expenses"                 → Income Statement → Operating Expenses
"P&L Depreciation"                   → Income Statement → Operating Expenses
```

2.4 Account Hierarchy (Parent Group)

Accounts are organized in a hierarchy that drives how reports group and subtotal balances:

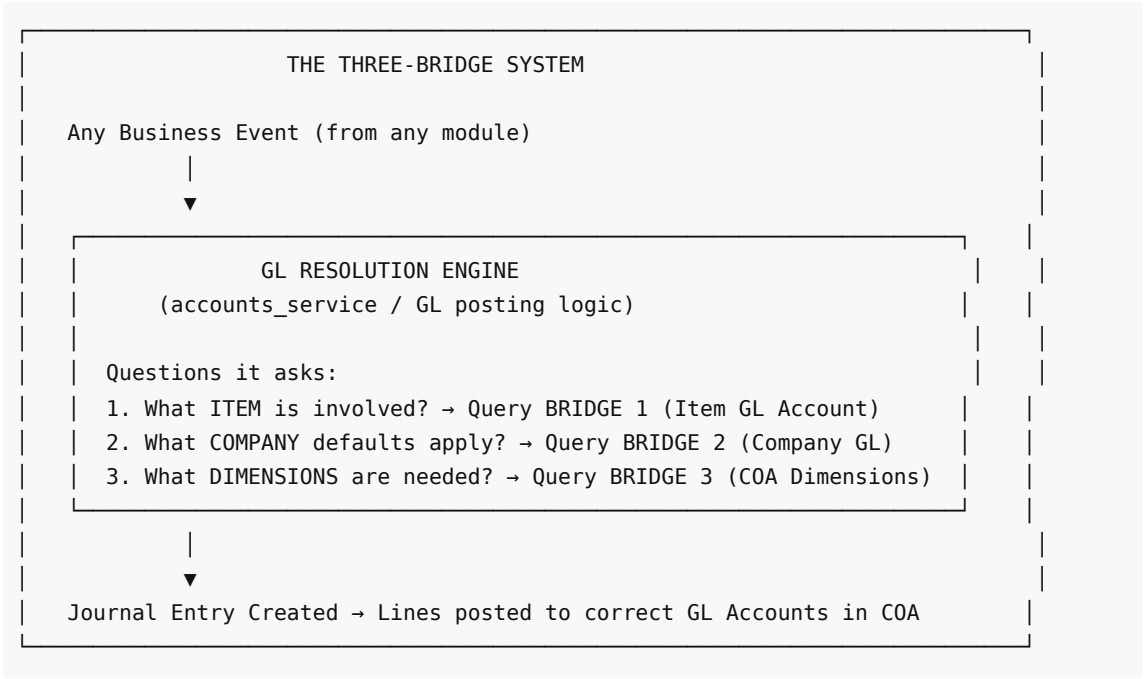
```
COA Hierarchy (Example for Assets)
│
├── BS Assets (Root Group)
│       │
│       ├── BS Cash Banks (Parent Group)
│       │       ├── Cash on Hand — Account Code: 1001001  [DEBIT]
│       │       ├── Petty Cash   — Account Code: 1001002  [DEBIT]
│       │       └── Main Bank Account — Account Code: 1001100  [DEBIT]
│       │
│       ├── BS Inventories (Parent Group)
│       │       ├── Finished Goods — Account Code: 2010400  [DEBIT]
│       │       ├── Raw Materials  — Account Code: 2010100  [DEBIT]
│       │       └── WIP            — Account Code: 2010200  [DEBIT]
│       │
│       └── BS Trade Receivables (Parent Group)
│               ├── Customer A Receivable — Account Code: 3010001  [DEBIT]
│               └── General Receivable    — Account Code: 3010000  [DEBIT]
```

---

3. The Three-Bridge System — How COA Connects to Every Module

The COA Engine does not connect directly to modules. Instead, it uses **three configuration bridges** that act as the translation layer between business events and GL accounts. This is the key to the

dynamic and flexible design.



Bridge 1 — Item GL Account (Product-Level Routing)

This is the **item-specific GL mapping**. When a transaction involves a product or service, the system looks up this bridge to know which accounts to use.

```
ITEM GL ACCOUNT BRIDGE
│
├── Main GL Account          → Primary stock/asset account
├── Sales Revenue Account    → Where sales income goes (P&L Income)
├── Trade Discounts Account  → Where discounts are recorded (P&L contra)
├── Cost of Goods Sold       → Where COGS goes on sale (P&L Expense)
├── Deferred Expenses Account→ For deferred cost items (BS Asset)
├── Output VAT Account       → Output VAT GL (BS Liability)
├── Input VAT Account        → Input VAT GL (BS Asset)
└── Goods on Consignment     → For consignment-based inventory (BS Asset)
```

How it works dynamically: When a Sales Invoice is posted for Item X:

- System looks up Item X's `ItemGLAccount`
- Gets `salesRevenueAccount` → posts Credit to that GL
- Gets `costOfGoodsSoldAccount` → posts Debit to that GL
- Gets `outputVATAccount` → posts Credit for VAT portion

This means each item can have a completely different set of GL accounts. Product A might post to "Sales of Finished Goods" while Service B posts to "Sales of Services" — all automatically resolved.

Bridge 2 — Company GL Account (Partner-Level Routing)

This bridge handles the **default GL accounts for receivables, payables, and payment methods** — the accounts that depend on the company (business entity) rather than the item.

```
COMPANY GL ACCOUNT BRIDGE
│
├── CUSTOMER-SIDE ACCOUNTS
│       ├── Accounts Receivable (Trade Receivables)
│       ├── Advances Received (Prepayments from Customers)
│       ├── PDCs Received (Post-Dated Cheques from Customers)
│       ├── Contract Assets (Unbilled Revenue)
│       ├── Retention Receivables
│       └── Retention Output VAT
│
├── SUPPLIER-SIDE ACCOUNTS
│       ├── Accounts Payable (Trade Payables)
│       ├── Advances Paid (Prepayments to Suppliers)
│       ├── PDCs Issued (Post-Dated Cheques to Suppliers)
│       ├── Retention Payables
│       ├── Retention Input VAT
│       └── Unbilled Purchases
│
└── PAYMENT & CASH ACCOUNTS
        ├── Cash Account (Cash on Hand)
        ├── Bank Account (Bank balance GL)
        └── Cheque Account (PDC clearing)
```

Why this is separate from Item GL? Because a company may deal with many different items, but the accounts receivable, payable, and payment method accounts are always the same regardless of what item was sold or purchased. Company A always uses GL `3010001` for receivables from their customers, regardless of what product was invoiced.

Bridge 3 — Account Dimensions (Analytical Routing)

This bridge controls how the COA Engine **breaks down balances for deeper analysis**. Instead of just posting to "Goods - AED 50,000", it can post to "Goods - Item: Steel Wire Coil - Warehouse: Main Store - AED 50,000".

```
DIMENSION SYSTEM
│
├── Available Dimension Types (25 types):
│       ├── ITEM                  → Break down by product/SKU
│       ├── WAREHOUSE             → Break down by storage location
│       ├── FIXED_ASSET           → Track per physical asset
│       ├── EMPLOYEE              → Break down by staff member
│       ├── TAX_TYPE              → Separate VAT types in one account
│       ├── COMPANY               → Track per business entity
│       ├── CONTRACT              → Break down by project/contract
│       ├── BUSINESS_ACTIVITY     → By line of business
│       ├── CASH_FLOW_ITEM        → Tag for cash flow statement
│       ├── EXPENSE_ITEM          → Expense category breakdown
│       ├── INCOME_ITEM           → Income category breakdown
│       ├── BANK_ACCOUNT          → Per bank account tracking
│       ├── DEFERRED_EXPENSE      → Per deferral schedule
│       ├── DEFERRED_REVENUE      → Per revenue recognition schedule
```

```
│       ├── EQUITY_MOVEMENTS_ITEMS→ Per equity change type
│       └── ... (10 more types)
│
└── Per Dimension, you configure:
        ├── Track Amount?         → Yes/No (track monetary value)
        ├── Track Quantitative?   → Yes/No (track physical units)
        └── Turnovers Only?       → Yes/No (only show movement, not balance)
```

Example — Goods Account with Two Dimensions:

```
Account: Finished Goods (GL: 2010400)
    Dimension 1: ITEM      → Track Amount + Quantity
    Dimension 2: WAREHOUSE → Track Amount only

Result in Trial Balance:
    Finished Goods — Total: AED 120,000
        ├── Item: Adult Hanger (1000pcs)   — AED 50,000
        │       ├── Warehouse: Main Store  — AED 35,000
        │       └── Warehouse: Branch      — AED 15,000
        ├── Item: Child Hanger (500pcs)    — AED 30,000
        └── Item: Heavy Duty (300pcs)      — AED 40,000
```

---

4. Dynamic GL Resolution Flow

This is the complete end-to-end flow of how the COA Engine resolves a GL account for any business event from any module:

```
┌──────────────────────────────────────────────────────────────────────────┐
│                    DYNAMIC GL RESOLUTION FLOW                            │
└──────────────────────────────────────────────────────────────────────────┘


STEP 1 — Event Raised in Source Module
──────────────────────────────────────

    Any Module (Sales, Purchases, Warehouse, HR, Fleet, etc.)
    raises a business event:
    "Sales Invoice #INV-2025-001 — Tenant: CompanyA — POSTED"
                │
                ▼

STEP 2 — GL Resolution Engine Receives the Event
─────────────────────────────────────────────────

    The accounts_service layer receives:
    ├── Document Type: CUSTOMER_INVOICE
    ├── Invoice Lines: [{item: Steel Wire, qty: 100, price: 500}]
    ├── VAT Rate: 5%
    ├── Customer: Supplier Co. Ltd
    └── Payment method: BANK
                │
                ▼

STEP 3 — Three-Bridge Lookup
```

```
──────────────────────────────────

    For each invoice line:

    BRIDGE 1 (Item GL): Look up Item "Steel Wire"
        → salesRevenueAccount   = GL 4010001 (Sales of Goods)
        → costOfGoodsSoldAccount = GL 5010001 (COS: Goods for Resale)
        → outputVATAccount      = GL 3030001 (Current VAT Payable)

    BRIDGE 2 (Company GL): Look up Company defaults
        → accountsReceivable    = GL 3010001 (Trade Receivables)
        → cashAccount / bankAccount = GL 1001100 (Bank Account)

    BRIDGE 3 (Dimensions): Check account dimensions
        → GL 4010001 has INCOME_ITEM dimension → tag with IncomeItem
        → GL 5010001 has ITEM dimension       → tag with item_id
                │
                ▼

STEP 4 — Build Journal Entry
──────────────────────────────────

    Journal Entry: JE-2025-10045
    ├── Line 1: Dr  Trade Receivables   GL:3010001  AED 5,250.00
    ├── Line 2: Cr  Sales of Goods      GL:4010001  AED 5,000.00  [dim: income_item]
    ├── Line 3: Cr  Current VAT Payable GL:3030001  AED   250.00  [dim: tax_type]
    └── Validation: Total Dr (5,250) = Total Cr (5,000 + 250) ✅ BALANCED
                │
                ▼

STEP 5 — Post to COA
──────────────────────────

    journal_entries table: entry saved, status = POSTED
    journal_entry_lines table: 3 lines saved with GL account references
    COA account balances: immediately updated in reporting queries
                │
                ▼

STEP 6 — Financial Statements Reflect Instantly
──────────────────────────────────────────────────

    Balance Sheet:
        Trade Receivables ↑ AED 5,250
        Current VAT Payable ↑ AED 250

    Income Statement:
        Sales of Goods ↑ AED 5,000
```

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5. Dimension System — Analytical Layer

5.1 Why Dimensions Exist

Without dimensions, the COA only tells you **how much** is in an account. With dimensions, it tells you **what, where, and by whom** that amount belongs to. This is the bridge between simple accounting

and **management accounting**.

5.2 Dimension Flow — How Dimensions Get Tagged

```
Business Event
    │
    ├── Source module passes context:
    │       ├── item_id (which product)
    │       ├── warehouse (which location)
    │       ├── employee_id (who)
    │       └── contract_id (which project)
    │
    ▼
GL Resolution Engine checks: "Does this GL account have dimensions configured?"
    │
    ├── YES → attach dimension references to the journal_entry_line
    └── NO  → post without dimension tagging
```

5.3 What Each Dimension Enables

Dimension	What It Enables	Which Module Uses It
ITEM	Stock value per product, COGS per SKU	Purchases, Sales, Warehouse, Production
WAREHOUSE	Inventory value per location	Warehouse, Production, Hanger Mfg
FIXED_ASSET	Depreciation and NBV per asset	Fixed Assets
EMPLOYEE	Salary cost per person	HR/Payroll
CONTRACT	Revenue and cost per project	Project Management, Sales
CASH_FLOW_ITEM	Classify transactions for Cash Flow Statement	All payment-related modules
EXPENSE_ITEM	Break down operating expenses by category	Purchases, Fleet, HR
INCOME_ITEM	Break down revenue by category	Sales, CRM, Fleet
TAX_TYPE	VAT analysis by tax type	VAT Module
BANK_ACCOUNT	Cash position per bank account	Bank Management
DEFERRED_EXPENSE	Track recognition per deferral record	Deferred Expenses
DEFERRED_REVENUE	Track recognition per revenue record	Deferred Revenue

---

6. Balance Sheet (BS) — Deep Dive

The Balance Sheet in this system is divided into three pillars: **Assets**, **Liabilities**, and **Equity**. Every GL account in the COA belongs to exactly one BS section (or to the Income Statement). Here is the complete deep explanation.

6.1 ASSETS

Assets are what the company **owns or controls**. All asset accounts have a **DEBIT** normal balance — they increase with debits and decrease with credits.

6.1.1 Current Assets

Current assets are expected to be converted to cash or used within one year.

🏦 BS Cash & BS Cash Banks

```
What it represents:
    Physical cash held in the office (cash on hand) and balances
    maintained in bank accounts.

GL accounts typically under this section:
    ├── Cash on Hand              ← Petty cash, register cash
    └── Bank Account (per bank)  ← Current account, savings account

What feeds it:
    ├── Payment receipts from customers (Dr Bank / Cr Receivables)
    ├── Payment to suppliers (Dr Payables / Cr Bank)
    ├── Payroll payments (Dr Wages Payable / Cr Bank)
    └── All cash/bank transactions from any module

What it shows in the BS:
    The total liquid cash available to the company on the BS date.

Dimension: BANK_ACCOUNT
    → So the BS can show "Total Cash: AED 250,000"
    → And drill down: "Bank A: AED 180,000 | Bank B: AED 50,000 | Cash: AED 20,000"
```

💰 BS Cash Equivalents

```
What it represents:
    Short-term, highly liquid investments that are easily convertible
    to known cash amounts (e.g., Treasury Bills, Money Market funds,
    short-term bank deposits < 3 months).

    "Cash Equivalents are items that are similar to cash such as
    short-time investment securities, that can be cashed out rather
    easily and quickly."  — Accounting Module Spec

What feeds it:
    When the company places a short-term investment:
        Dr: Cash Equivalent account (BS Cash Equivalents)
        Cr: Bank Account (BS Cash)
```

```
What separates it from BS Cash:
    Cash is immediately available. Cash Equivalents are liquid but
    require a transaction to convert (e.g., selling a T-Bill).

    BS shows both as "Liquid Assets" but keeps them separate for
    IFRS disclosure requirements.
```

📄 BS Current Financial Assets

```
What it represents:
    Short-term financial instruments: short-term receivables from
    financial activities, short-term investments in securities,
    short-term loans given to third parties.

What feeds it:
    ├── Short-term investment securities purchases
    ├── Loans given to employees / partners (short-term)
    └── Financial instrument transactions

Distinguished from Cash Equivalents:
    Financial Assets have a slightly longer horizon or involve
    a specific financial instrument contract. Cash Equivalents
    are freely tradeable in < 3 months.
```

📦 BS Inventories

```
What it represents:
    The value of all goods held for sale, raw materials for
    production, and work-in-progress at any point in time.

GL accounts typically under this section:
    ├── Goods (Finished goods for resale)  ← Primary trading inventory
    ├── Raw Materials                       ← Steel wire coils (hanger client)
    ├── Work In Progress (WIP)              ← Partially manufactured goods
    └── Packaging Materials                ← Boxes, labels, tape

What feeds it:
    ├── Purchase Invoice Posted:
    │       Dr Goods/Raw Materials (BS Inventories) ← INCREASES balance
    │       Cr Current Trade Payables
    │
    ├── Sales Invoice Posted (goods delivered):
    │       Dr COS: Goods for Resale (P&L Expense) ← REDUCES inventory
    │       Cr Goods (BS Inventories)
    │
    ├── Inventory Transfer (between warehouses):
    │       Dr Goods – Destination WH (BS Inventories)
    │       Cr Goods – Source WH (BS Inventories)
    │       (Net BS impact = zero, but warehouse dimension changes)
    │
    ├── Inventory Write-Off:
```

```
    │       Dr Other Operating Expenses (P&L)
    │       Cr Goods (BS Inventories)  ← REDUCES balance
    │
    └── Production Material Issue:
            Dr WIP (BS Inventories)
            Cr Raw Materials (BS Inventories)

Dimensions configured:
    ├── ITEM      → Know the value of each product in inventory
    └── WAREHOUSE → Know how much stock is at each location

Quantitative tracking enabled:
    → Track both AED value AND physical quantity (pcs, kg, etc.)
    → Allows: "Steel Wire Coil: 500 Kg remaining = AED 12,500"
```

🏢 BS Trade Receivables (Current)

```
What it represents:
    Money owed TO the company by customers for goods/services
    already delivered but not yet paid for.

What feeds it:
    ├── Sales Invoice Posted:
    │       Dr Trade Receivables ← INCREASES balance
    │       Cr Sales of Goods / Services
    │
    ├── Customer Payment Received:
    │       Dr Bank/Cash ← increases cash
    │       Cr Trade Receivables ← REDUCES balance
    │
    ├── Credit Note Issued to Customer:
    │       Dr Sales of Goods (reversal) / Operating Expenses
    │       Cr Trade Receivables ← REDUCES balance
    │
    └── Customer Advance applied to Invoice:
            Dr Customer Advance (contra)
            Cr Trade Receivables ← REDUCES balance

Default GL is configured in:
    CompanyGLAccount.accountsReceivable
    → Each company has its own "Trade Receivables" GL account

Dimension: Typically tracked per COMPANY (customer-wise)
    → "Total Receivables: AED 500,000"
    → Drill: "Customer A: AED 200,000 | Customer B: AED 300,000"
```

💳 BS Advances Paid / Prepayments to Suppliers

```
What it represents:
    Money the company has paid to suppliers in advance BEFORE
    receiving goods or services. This is an asset because the
```

```
    company is owed delivery in the future.

What feeds it:
    ├── Advance Payment to Supplier:
    │       Dr Prepayments to Suppliers ← ASSET created
    │       Cr Bank / Cash
    │
    └── Invoice Received against advance:
            Dr Goods / Expense
            Cr Prepayments to Suppliers ← ASSET consumed
            Dr/Cr difference → Current Trade Payables (any balance)

Configured in: CompanyGLAccount.advancesPaid
```

📋 BS PDCs Received from Companies

```
What it represents:
    Post-dated cheques received from customers. These are assets
    because they represent a promise of future payment. They sit
    as receivables until the cheque's date arrives and it clears.

Two-step lifecycle:
    Step 1 — Cheque received:
        Dr PDCs Received from Companies (BS Asset) ← Created
        Cr Trade Receivables ← Customer balance cleared

    Step 2 — Cheque clearing date (cheque deposited):
        Dr Bank Account (BS Cash)
        Cr PDCs Received from Companies ← Asset consumed

Configured in: CompanyGLAccount.pdcsReceived
```

🧾 BS Input VAT (Current Tax Receivable)

```
What it represents:
    VAT paid on purchases that the company can claim back from
    the tax authority (UAE: FTA). This is a recoverable amount —
    hence it is an ASSET.

What feeds it:
    Every purchase invoice with VAT:
        Dr Goods                    (net amount)
        Dr Current VAT Receivable   (5% of net) ← ASSET accumulates
        Cr Current Trade Payables   (gross amount)

What reduces it:
    VAT Settlement at period end:
        Dr Current VAT Payable      (offset against liability)
        Cr Current VAT Receivable   ← net off against output VAT

    If Input VAT > Output VAT → Company receives a refund from FTA
```

```
    If Output VAT > Input VAT → Company pays the difference to FTA

This is separate from "BS Current Other Tax Receivable" which
covers other types of taxes (income tax, municipal tax, etc.)
```

📅 **BS Deferred Expenses (Current & Non-Current)**

```
What it represents:
    Expenses already PAID but not yet "consumed" from a business
    perspective. Example: Annual insurance premium paid upfront —
    only 1/12 is expense each month, the rest stays as an asset.

Two sub-sections:
    BS Deferred Expenses          → Will be recognized within 1 year (Current)
    BS Non-Current Deferred Expenses → Will be recognized after 1 year

What feeds it:
    Purchase invoice with deferred expense item:
        Dr Deferred Expenses (BS Asset) ← Created as asset
        Cr Current Trade Payables

Each month as it is recognized:
    Dr Admin/Operating Expenses (P&L) ← Expense hits P&L
    Cr Deferred Expenses (BS Asset)  ← Asset reduces

Recognition methods: Monthly / Daily / Manual
```

📑 **BS Contract Assets**

```
What it represents:
    Revenue earned but NOT YET BILLED to the customer (unbilled
    revenue). This typically arises in long-term contracts where
    work has been completed but the invoice has not been raised yet.

    Common in: Project Management, Construction, Long-term Service contracts

What feeds it:
    When revenue is recognized on a partially complete milestone
    but the invoice has not been issued yet.

Distinguished from Trade Receivables:
    Trade Receivables = Invoice has been raised and customer owes money
    Contract Assets   = Right to receive money exists but invoice not yet raised
```

6.1.2 Non-Current Assets

Non-current assets are long-term assets not expected to convert to cash within one year.

🏭 **BS Fixed Assets (Property, Plant & Equipment)**

```
What it represents:
    Tangible long-term assets used in operations:
    Buildings, machinery, vehicles, computers, furniture.

Sub-sections within Fixed Assets:
    ├── BS Fixed Assets Acquired      ← Original cost of PPE
    ├── BS Capital Improvements to PPE ← Subsequent improvements
    └── BS Assets Classified as Held for Sale - Fixed Assets ← Being sold

What feeds it:
    Asset Purchase:
        Dr Fixed Assets Acquired (BS) ← Asset created at cost
        Cr Bank / Trade Payables

    Depreciation (monthly):
        Dr Depreciation Expense (P&L) ← Expense recognized
        Cr Accumulated Depreciation   ← Contra asset (reduces net book value)

    Asset Disposal:
        Dr Accumulated Depreciation
        Dr Loss on Disposal (P&L) or Cr Gain on Disposal (P&L)
        Cr Fixed Assets Acquired      ← Asset removed from BS

Net Book Value shown in BS:
    Fixed Assets – Cost: AED 500,000
    Less: Accumulated Depreciation: AED (120,000)
    Net Book Value: AED 380,000
```

🏗️ **BS Intangible Assets & BS Investment Property**

```
BS Intangible Assets:
    Non-physical long-term assets: software licenses, patents,
    trademarks, goodwill.
    ├── BS Intangible Assets
    ├── BS Capital Improvements to Intangible Assets
    └── BS Construction in Progress of Intangible Assets

BS Investment Property:
    Properties held for rental income or capital appreciation
    (NOT for owner-occupation).
    ├── BS Investment Property
    ├── BS Capital Improvements to Investment Property
    └── BS Construction in Progress of Investment Property
```

6.1.3 Summary — Assets Structure in Balance Sheet

```
ASSETS (Balance Sheet)
│
├── CURRENT ASSETS
│       ├── Cash and Cash Equivalents
```

```
│       │       ├── BS Cash (Cash on Hand + Bank Accounts)
│       │       └── BS Cash Equivalents (Short-term liquid investments)
│       │
│       ├── BS Current Financial Assets (Short-term investments/loans)
│       ├── BS Inventories (Goods, Raw Materials, WIP)
│       ├── BS Trade Receivables (Customer balances)
│       ├── BS Input VAT (Recoverable VAT)
│       ├── BS Contract Assets (Unbilled revenue)
│       ├── BS Deferred Expenses (Prepaid costs < 1 year)
│       ├── BS Prepayments to Suppliers (Advance payments)
│       └── BS PDCs Received (Cheques from customers)
│
└── NON-CURRENT ASSETS
        ├── BS Fixed Assets (PPE — net of depreciation)
        ├── BS Capital Improvements to PPE
        ├── BS Intangible Assets (Software, Patents, Goodwill)
        ├── BS Investment Property (Properties for rental)
        ├── BS Non-Current Financial Assets (Long-term investments)
        ├── BS Non-Current Deferred Expenses (> 1 year prepaid costs)
        ├── BS Non-Current Contract Assets (Long-term unbilled revenue)
        ├── BS Deferred Tax Assets
        └── BS Construction in Progress (Projects under construction)
```

6.2 LIABILITIES

Liabilities are what the company **owes to others**. All liability accounts have a **CREDIT** normal balance — they increase with credits and decrease with debits.

6.2.1 Current Liabilities

💸 BS Current Trade Payables

```
What it represents:
    Money owed BY the company to suppliers for goods/services
    already received but not yet paid for.

What feeds it:
    ├── Purchase Invoice Posted:
    │       Dr Goods / Expenses / Input VAT
    │       Cr Current Trade Payables ← LIABILITY created
    │
    ├── Supplier Payment Made:
    │       Dr Current Trade Payables ← LIABILITY reduced
    │       Cr Bank / Cash / PDC Issued
    │
    └── Return of Goods to Supplier:
            Reversal of original invoice entries
            → LIABILITY reduced or offset with prepayment

Configured in: CompanyGLAccount.accountsPayable
```

🧾 BS Current VAT Payable (Output VAT)

```
What it represents:
    VAT collected FROM customers on sales that must be remitted
    to the UAE Federal Tax Authority (FTA). This is a liability
    because it does not belong to the company — it was collected
    on behalf of the government.

What feeds it:
    Every sales invoice with VAT:
        Dr Trade Receivables    (gross with VAT)
        Cr Sales of Goods       (net)
        Cr Current VAT Payable  (5% of net) ← LIABILITY grows

What reduces it:
    VAT Settlement payment to FTA:
        Dr Current VAT Payable ← LIABILITY cleared
        Cr Bank Account

    Or netted against Input VAT:
        Dr Current VAT Payable (output)
        Cr Current VAT Receivable (input) if input > output
```

💳 BS Advances from Customers (Prepayments)

```
What it represents:
    Cash received FROM customers BEFORE goods/services are
    delivered. The company owes delivery — hence it's a LIABILITY.

What feeds it:
    Customer makes a prepayment:
        Dr Bank / Cash / PDC
        Cr Advances from Customers ← LIABILITY

When goods are delivered and invoice raised:
    Dr Advances from Customers ← LIABILITY cleared
    Cr Sales of Goods / Services

Configured in: CompanyGLAccount.advancesReceived
```

📋 BS PDCs Issued to Companies

```
What it represents:
    Post-dated cheques issued TO suppliers. These are liabilities
    because the company has committed to future payments.

Two-step lifecycle:
    Step 1 — Cheque issued:
        Dr Current Trade Payables ← Supplier balance cleared
        Cr PDCs Issued to Companies ← LIABILITY created

    Step 2 — On clearing date:
        Dr PDCs Issued to Companies ← LIABILITY cleared
```

```
        Cr Bank Account

    If bank charges on clearing:
        Additional Dr Bank Charges / Cr Bank Account

Configured in: CompanyGLAccount.pdcsIssued
```

📅 BS Deferred Revenue (Current & Non-Current)

```
What it represents:
    Revenue received in cash or invoiced but NOT YET EARNED
    because the service has not been delivered yet.
    Example: Annual subscription fee collected upfront.

    BS Deferred Revenue          → Portion to be earned within 1 year
    BS Non-Current Deferred Revenue → Portion to be earned after 1 year

What feeds it:
    Service invoice raised with deferred option:
        Dr Trade Receivables (or Cash)
        Cr Deferred Revenue ← LIABILITY (revenue not yet earned)

As service is delivered over time:
    Dr Deferred Revenue ← LIABILITY reduces
    Cr Sales of Services (P&L) ← Income recognized

Recognition methods: Monthly / Daily / Manual
```

🧮 Other Current Liabilities

```
BS Contract Liabilities
    → Customer payments received for performance obligations not yet fulfilled
    → Differs from Deferred Revenue in IFRS 15 treatment

BS Non-Current Contract Liabilities
    → Same but for obligations beyond 1 year

BS Current Income Tax Payable
    → Corporate income tax owed to government for current period

BS Current Other Tax Payable
    → Municipal taxes, other non-VAT, non-income taxes

BS Current Financial Liabilities
    → Short-term bank loans, overdrafts, current portion of long-term loans
```

6.2.2 Non-Current Liabilities

```
BS Non-Current Financial Liabilities
    → Long-term bank loans, bonds payable, lease liabilities > 1 year
```

```
BS Deferred Tax Liabilities
    → Future tax obligations arising from temporary accounting differences

BS Non-Current Contract Liabilities
    → Long-term service obligations

BS Non-Current Deferred Revenue
    → Revenue deferred beyond 1 year
```

6.3 EQUITY

Equity represents the **owners' residual interest** in the company after all liabilities are subtracted. All equity accounts have a **CREDIT** normal balance.

```
EQUITY (Balance Sheet)
│
├── BS Capital Issued                    ← Share capital / owner's investment
│       (Issued Share Capital, Share Premium, Treasury Shares)
│       Managed through: EquityAccount entity + EquityTransaction records
│
├── Retained Earnings                    ← Accumulated profits not distributed
│       = Opening Retained Earnings + Net Profit (from P&L) − Dividends
│       The P&L surplus/deficit AUTOMATICALLY flows into Retained Earnings
│       at year-end through the closing entry
│
└── BS Non-Controlling Interests         ← Equity held by minority shareholders
        (Relevant for consolidated reporting)
```

The Golden Equation:

```
Assets = Liabilities + Equity

Every journal entry maintains this equation:
    When Dr = Cr, the equation always balances.
    No transaction can ever break Assets = Liabilities + Equity.
```

---

7. Income Statement (P&L) — Deep Dive

The Income Statement measures **financial performance** over a period (month, quarter, year). Unlike the Balance Sheet (which is a snapshot), the P&L is a flow statement — it accumulates transactions over a period.

Structure:

```
Revenue
  − Cost of Sales
  ──────────────
  = Gross Profit

  − Operating Expenses
  ────────────────────
```

```
  = Operating Profit (EBIT)

  ± Finance Income/Costs
  ──────────────────────
  = Net Profit Before Tax

  − Income Tax
  ────────────
  = Net Profit After Tax
```

7.1 REVENUE (Income)

Revenue accounts have a **CREDIT** normal balance — they increase with credits and decrease with debits.

🛒 Sales of Goods

```
What it represents:
    Revenue earned from selling physical products (inventory items).

What feeds it:
    Every Sales Invoice for an inventory item:
        Dr Trade Receivables (gross)
        Cr Sales of Goods ← INCOME grows with every sale

    Credit Notes / Returns reduce it:
        Dr Sales of Goods ← INCOME reversed
        Cr Trade Receivables / Prepayment

GL Account configured in:
    ItemGLAccount.salesRevenueAccount (per item)
    → Different product lines can map to different revenue accounts:
    → "Sales of Finished Goods" vs "Sales of Raw Materials"

Modules that feed this:
    ├── Sales Module (direct customer invoices)
    ├── POS Module (point-of-sale receipts)
    ├── CRM Module (rental quotation invoices)
    └── Production Module (if manufacturing sells direct)
```

🔧 Sales of Services

```
What it represents:
    Revenue earned from delivering services (labour, consulting,
    maintenance, project-based work).

What feeds it:
    Service invoices (Item Type = SERVICE):
        Dr Trade Receivables
        Cr Sales of Services ← INCOME
```

```
    If service has deferred revenue:
        Initially: Cr Deferred Revenue (liability, not income)
        Monthly:   Dr Deferred Revenue → Cr Sales of Services

GL Account configured in:
    ItemGLAccount.salesRevenueAccount where item.itemType = SERVICE

Modules that feed this:
    ├── Sales Module (service invoices)
    ├── Fleet Module (trip/freight charges)
    ├── Project Management (milestone billing)
    └── ATS Module (upcoming: recruitment service fees)
```

📊 **Other Operating Income**

```
What it represents:
    Income from activities that are secondary to the main business:
    ├── Inventory adjustments (physical count gains)
    │       Inventory Receive: Dr Goods / Cr Other Operating Income
    ├── Scrap sales (hanger manufacturing)
    ├── Insurance claim receipts
    └── Miscellaneous receipts

IncomeItem dimension:
    Each credit to "Other Operating Income" is tagged with an
    INCOME_ITEM dimension so management can see breakdowns:
    "Other Income: AED 15,000
        ├── Inventory Count Gain: AED 5,000
        ├── Scrap Sales: AED 7,000
        └── Insurance Proceeds: AED 3,000"
```

7.2 COST OF SALES (COS)

Cost of Sales accounts have a **DEBIT** normal balance. They represent the direct cost of producing or acquiring the goods/services sold.

📦 **COS: Goods for Resale**

```
What it represents:
    The COST of inventory items sold to customers. When a product
    is sold, its cost moves from the Balance Sheet (BS Inventories)
    to the Income Statement (COS).

What feeds it:
    Sales Invoice Posted (with goods delivery):
        Dr COS: Goods for Resale ← P&L COST increases
        Cr Goods (BS Inventories) ← BS ASSET decreases

This is the fundamental double movement:
    → The asset (inventory) converts into an expense (COGS)
    → The difference between Sales Revenue and COGS = Gross Margin
```

```
GL Account configured in:
    ItemGLAccount.costOfGoodsSoldAccount (per item)

Modules that feed this:
    Sales Module, POS Module, Production Module
```

7.3 OPERATING EXPENSES

Operating expense accounts have a **DEBIT** normal balance.

🏢 **Other Administrative Expenses**

```
What it represents:
    Overhead costs of running the business that are not directly
    tied to production or sales:
    ├── Office supplies and stationery
    ├── Utilities (electricity, water, internet)
    ├── Office rent
    ├── Professional services (audit, legal)
    └── General administrative costs

What feeds it:
    Expense invoices (non-inventory, non-COGS items):
        Dr Other Administrative Expenses ← P&L EXPENSE
        Cr Current Trade Payables

ExpenseItem dimension:
    "Admin Expenses: AED 50,000
        ├── Office Rent: AED 25,000
        ├── Utilities: AED 10,000
        └── Professional Fees: AED 15,000"

ResourceType filter in ExpenseItem:
    ALL / INVENTORY_ITEM / FIXED_ASSET / MANPOWER / SERVICE
    → So expense categories are filtered by what type of resource
      is being expensed
```

🔧 **Other Operating Expenses**

```
What it represents:
    Costs directly related to operations but not classified as
    administrative:
    ├── Inventory write-offs
    │       Dr Other Operating Expenses ← EXPENSE
    │       Cr Goods (BS Inventories)
    │
    ├── Credit note adjustments to customers
    │       Dr Other Operating Expenses ← EXPENSE
    │       Cr Trade Receivables
    │
```

```
    └── Operational losses and provisions

Distinction from Administrative Expenses:
    Administrative = General overhead (runs regardless of sales volume)
    Operating      = Directly operational (varies with operational activity)
```

📉 Trade Discounts and Rebates on Sales

```
What it represents:
    Discounts given to customers that reduce net revenue.
    This is a CONTRA-REVENUE account — it offsets Sales of Goods.

What feeds it:
    Sales Invoice with discount:
        Dr Trade Receivables (net of discount)
        Dr Trade Discounts & Rebates ← CONTRA reduces revenue
        Cr Sales of Goods (gross)

P&L Presentation:
    Sales of Goods:              AED 100,000
    Less: Trade Discounts:      AED (5,000)
    Net Revenue:                 AED  95,000

GL Account configured in:
    ItemGLAccount.tradeDiscountsAccount (per item)
```

💼 Depreciation Expense

```
What it represents:
    The systematic allocation of the cost of a fixed asset over
    its useful life. This is NOT a cash payment — it is an
    accounting entry to recognize wear and tear.

What feeds it (monthly, automated from Fixed Assets module):
    Dr Depreciation Expense (P&L) ← EXPENSE recognized
    Cr Accumulated Depreciation (BS contra-asset)

This is why EBITDA (Earnings Before Interest, Tax, Depreciation,
Amortization) is important — it shows profit before this
non-cash charge.

Modules that generate this:
    Fixed Assets Module (depreciation schedules run monthly)
```

👥 Payroll & Staff Costs

```
What it represents:
    All employee-related costs:
    ├── Basic salaries
    ├── Allowances (housing, transport)
    ├── End of Service Benefits (EOSB) provision
```

```
    └── Social insurance contributions

What feeds it (from HR/Payroll Module):
    Payroll run posted:
        Dr Salaries Expense (P&L) ← EXPENSE
        Cr Salaries Payable (BS Liability)

    EOSB provision:
        Dr EOSB Expense (P&L) ← EXPENSE
        Cr EOSB Provision (BS Liability)

Employee dimension available:
    → So management can see cost per employee or department
```

⛽ **Fleet & Transport Expenses**

```
What it represents:
    Vehicle running costs recorded through the Fleet Module:
    ├── Fuel expenses
    ├── Vehicle maintenance & repairs
    └── Road tax and vehicle registration

What feeds it:
    Fleet Module expense entries:
        Dr Fuel Expense / Vehicle Maintenance (P&L)
        Cr Cash / Trade Payables
```

🏗️ **Production / Manufacturing Costs**

```
What it represents:
    Production overheads and processing costs:
    ├── Machine operating costs
    ├── Factory utilities
    ├── Production labour (in addition to payroll)
    └── Scrap and wastage

What feeds it:
    Production Module / Hanger Manufacturing Module:
        Dr Production Overheads (P&L)
        Cr Various payables / accruals

    Scrap from production:
        Dr Scrap Expense (P&L)
        Cr WIP (BS Inventories)
```

7.4 Summary — Income Statement Structure

```
INCOME STATEMENT (P&L)
│
├── REVENUE
```

```
│       ├── Sales of Goods             ← From Sales, POS, CRM modules
│       ├── Sales of Services          ← From Sales, Fleet, Project, ATS modules
│       ├── Other Operating Income     ← Inventory gains, misc income
│       └── Less: Trade Discounts & Rebates ← Customer discounts
│
├── COST OF SALES
│       └── COS: Goods for Resale     ← COGS when inventory sold
│
│       ═══════════════════════════════
│       = GROSS PROFIT
│       ═══════════════════════════════
│
├── OPERATING EXPENSES
│       ├── Other Administrative Expenses ← Office, utilities, professional fees
│       ├── Other Operating Expenses      ← Write-offs, adjustments
│       ├── Depreciation Expense          ← From Fixed Assets module
│       ├── Salaries & Staff Costs        ← From HR/Payroll module
│       ├── Fleet & Transport             ← From Fleet module
│       └── Production Overheads         ← From Production/Hanger module
│
│       ═══════════════════════════════
│       = OPERATING PROFIT (EBIT)
│       ═══════════════════════════════
│
├── FINANCE INCOME / (COSTS)
│       ├── Interest Income              ← From Cash Equivalents investments
│       └── Interest Expense            ← Bank loan interest (Financial Liabilities)
│
│       ═══════════════════════════════
│       = NET PROFIT BEFORE TAX
│       ═══════════════════════════════
│
└── INCOME TAX EXPENSE
        └── Corporate Income Tax        ← Tax on profits
        ═══════════════════════════════
        = NET PROFIT AFTER TAX
        ═══════════════════════════════
        → Transferred to Retained Earnings (BS Equity) at year-end
```

---

8. How Each Module Feeds the BS & P&L

This section shows exactly which BS sections and P&L lines each active module contributes to:

```
┌──────────────────────────────────────────────────────────────────────────────┐
│              MODULE → BALANCE SHEET & P&L IMPACT MATRIX                      │
├──────────────────────┬──────────────────────────────────────┬────────────────┤
│ MODULE               │ BALANCE SHEET SECTIONS AFFECTED      │ P&L LINES AFFECTED   │
├──────────────────────┼──────────────────────────────────────┼────────────────┤
│ PURCHASES            │ BS Inventories ↑                     │ (No direct P&L)      │
│                      │ BS Trade Payables ↑                  │ Unless expense item  │
│                      │ BS Input VAT ↑                       │ → Admin Expenses ↑   │
```

	BS Prepayments to Suppliers ↑	
SALES	BS Trade Receivables ↑ BS Inventories ↓ (goods delivered) BS VAT Payable ↑ BS Advances from Customers ↑ BS PDCs Received ↑	Sales of Goods ↑ COS: Goods ↑ Trade Discounts ↑ Sales of Services ↑
WAREHOUSE	BS Inventories (moves between WH) (Net effect: dimension changes only)	Operating Expenses ↑ (write-offs only)
FIXED ASSETS	BS Fixed Assets ↑ (acquisition) BS Fixed Assets ↓ (disposal) BS Accumulated Depreciation ↑	Depreciation ↑ Gain/Loss disposal
HR / PAYROLL	BS Salaries Payable ↑ BS EOSB Provision ↑ BS Bank ↓ (payments)	Salaries Expense ↑ EOSB Expense ↑
FLEET	BS Trade Payables ↑ (fuel/maint) BS Bank ↓ (payments) BS Trade Receivables ↑ (trip billing)	Fuel Expense ↑ Maintenance Exp ↑ Trip Revenue ↑
PRODUCTION / HANGER MFG	BS Inventories (Raw Mat → WIP) BS Inventories (WIP → Finished) BS Inventories ↓ (when sold)	Production Cost ↑ Scrap Expense ↑ COS ↑ on sale
PROJECT MGMT	BS Contract Assets ↑ BS Trade Receivables ↑ BS Contract Liabilities ↑	Project Revenue ↑ Project Costs ↑
POS	BS Cash ↑ (daily sales) BS Inventories ↓ (goods sold) BS VAT Payable ↑	Sales of Goods ↑ COS ↑
CRM	BS Trade Receivables ↑ (through rental invoices)	Rental Revenue ↑ Service Revenue ↑
VAT MODULE	BS Input VAT ↑/↓ BS VAT Payable ↑/↓ BS Bank ↓ (VAT payment to FTA)	(No P&L impact) Unless penalties → Operating Exp ↑

---

9. Upcoming Module Integration Design

This section explains how the COA Engine is designed to **automatically accommodate new modules** without any changes to the COA structure itself. This is the true power of the dynamic design.

9.1 Design Principle for New Module Integration

```
WHEN A NEW MODULE IS ADDED:
│
├── STEP 1: The new module's Liquibase context creates its tables
│
├── STEP 2: The module's business documents generate events
│           (invoice, payment, transfer, payroll run, etc.)
│
├── STEP 3: The accounts_service GL Resolution Engine handles the event
│           using the same three bridges:
│           ├── ItemGLAccount (if it involves items/services)
│           ├── CompanyGLAccount (if it involves partners)
│           └── COA Dimensions (for analytical tracking)
│
├── STEP 4: The journal entry hits the same COA table
│           → No new accounting tables needed
│           → No new financial statement logic needed
│
└── STEP 5: The new module's transactions AUTOMATICALLY appear in:
            ├── Trial Balance
            ├── General Ledger
            ├── Income Statement
            └── Balance Sheet
```

> ***Key Insight:*** *The COA Engine is module-agnostic. It only cares about GL account codes, Dr/Cr amounts, and dimensions. Any module that speaks this language integrates instantly.*

9.2 ATS (Applicant Tracking System) Module

```
MODULE: ATS (Recruitment & HR Applicant Tracking)
STATUS: Upcoming / Active Development

HOW IT CONNECTS TO COA ENGINE:
│
├── Revenue Side:
│       When recruitment agency invoices a client for placement:
│           Dr Trade Receivables (CompanyGLAccount.accountsReceivable)
│           Cr Sales of Services (ItemGLAccount.salesRevenueAccount)
│           Cr Current VAT Payable (ItemGLAccount.outputVATAccount)
│
├── Expense Side:
│       Job board advertising costs:
│           Dr Recruitment Advertising Expense (P&L)
│           Cr Current Trade Payables
│
│       Background verification fees:
│           Dr HR Admin Expenses (P&L)
│           Cr Current Trade Payables
│
└── New Dimensions Potentially Needed:
        ├── JOB_POSITION → Track cost per job role advertised
        └── CANDIDATE    → Track cost per candidate pipeline
```

```
BS Impact: Trade Receivables ↑, Trade Payables ↑, Bank ↓ (on payment)
P&L Impact: Recruitment Revenue ↑, Advertising Expense ↑
```

9.3 Supply & Tendering Module

```
MODULE: Supply & Tendering (Bidding for contracts, supply chain)
STATUS: Documented / Upcoming

HOW IT CONNECTS TO COA ENGINE:
│
├── Tender Bidding Costs:
│       Tender documentation fees, bid bond costs:
│           Dr Tendering Expenses (P&L Operating Expenses)
│           Cr Cash / Bank
│
├── Bid Bond / Performance Bond:
│       When company places a guarantee with client:
│           Dr Bid Bond Deposits (BS — Off-Balance or BS Other Assets)
│           Cr Bank Account
│
│       When bond is released:
│           Dr Bank Account
│           Cr Bid Bond Deposits
│
├── Supply Contracts Won:
│       Contract recognized as revenue over delivery:
│           Dr Trade Receivables / Contract Assets
│           Cr Sales of Goods / Services
│           → Feeds directly into existing Sales sub-ledger
│
└── New Dimensions Potentially Needed:
        ├── TENDER_ID → Track cost per tender submission
        └── CONTRACT  → Already exists — track P&L per contract won

BS Impact: Contract Assets ↑, Bid Bond Deposits ↑/↓
P&L Impact: Tendering Costs ↑, Contract Revenue ↑ (when won)
```

9.4 Trip Management Module

```
MODULE: Trip Management (Connected to Fleet)
STATUS: Documented / Active

HOW IT CONNECTS TO COA ENGINE:
│
├── Revenue — Freight/Trip Charges:
│       Customer invoice for completed trip:
│           Dr Trade Receivables
│           Cr Freight Revenue / Transport Income (P&L)
│           Cr Current VAT Payable
```

```
│
├── Expenses — Direct Trip Costs:
│       Fuel, toll, driver allowance, parking:
│           Dr Fuel & Transport Expenses (P&L)
│           Cr Cash / Trade Payables
│
├── Driver Salary/Advance:
│       Advance to driver:
│           Dr Driver Advance (BS Current Asset or Expense)
│           Cr Cash
│
│       Advance settled from payroll:
│           Dr Salaries Payable
│           Cr Driver Advance
│
└── New Dimensions Potentially Needed:
        ├── TRIP_ID    → Track P&L per trip (profitability per trip)
        └── VEHICLE_ID → Track cost/revenue per vehicle

BS Impact: Trade Receivables ↑, Cash ↓ (fuel/expenses), Driver Advance ↑/↓
P&L Impact: Freight Revenue ↑, Fuel Expense ↑, Driver Allowance ↑
```

9.5 Project Costing Module

```
MODULE: Project Management & Costing
STATUS: Active (project_management package exists)

HOW IT CONNECTS TO COA ENGINE:
│
├── Project Revenue:
│       Milestone billing to client:
│           Dr Trade Receivables / Contract Assets (BS)
│           Cr Sales of Services / Project Revenue (P&L)
│
│       Retention withheld by client:
│           Dr Retention Receivables (BS Asset)
│           Cr Sales of Services (P&L)
│
├── Project Costs — Materials:
│       Materials issued to project from warehouse:
│           Dr Project Material Cost (P&L)
│           Cr Goods (BS Inventories)
│
│       Materials purchased for project directly:
│           Dr Project Material Cost (P&L)
│           Cr Current Trade Payables
│
├── Project Costs — Labour:
│       Labour allocated from payroll to project:
│           Dr Project Labour Cost (P&L)
│           Cr Salaries Allocated (contra payroll account)
```

```
│
├── Project Costs — Subcontractor:
│       Subcontractor invoice received:
│           Dr Subcontractor Costs (P&L)
│           Cr Current Trade Payables
│
└── CONTRACT Dimension is the Key:
        Every project transaction tagged with CONTRACT_ID
        → Complete project P&L available:
            "Project Alpha Revenue: AED 500,000
             Project Alpha Materials: AED (150,000)
             Project Alpha Labour:    AED  (80,000)
             Project Alpha Sub:       AED  (70,000)
             Project Alpha Profit:    AED  200,000"

BS Impact: Contract Assets ↑, Retention Receivables ↑, Inventories ↓
P&L Impact: Project Revenue ↑, Project Costs ↑ (various lines)
```

9.6 Universal Biometric / HRMS Expansion

```
MODULE: Biometric Integration → Advanced Payroll
STATUS: Documented / Upcoming

HOW IT CONNECTS TO COA ENGINE:
│
├── Overtime Calculations → Additional Salary Expense:
│       Dr Overtime Expense (P&L)
│       Cr Salaries Payable (BS Liability)
│
├── Leave Encashment:
│       Dr Leave Encashment Expense (P&L)
│       Cr Salaries Payable / Cash
│
├── Absence Deductions:
│       Reduces Dr Salaries Expense
│       Reduces Cr Salaries Payable
│       (Salary run is net after deductions)
│
└── EMPLOYEE dimension enabled on salary accounts:
        → Management sees payroll cost per employee, per department

BS Impact: Salaries Payable ↑, Bank ↓ (on payment)
P&L Impact: Salaries ↑, Overtime ↑, EOSB Provision ↑
```

9.7 POS (Point of Sale) Module

```
MODULE: POS — Point of Sale
STATUS: Active

HOW IT CONNECTS TO COA ENGINE:
```

```
│
├── POS Sale Transaction:
│       Dr Cash on Hand (BS Cash) ← Immediate cash collection
│       Cr Sales of Goods (P&L)
│       Cr Current VAT Payable (BS Liability)
│
├── COGS on POS Sale:
│       Dr COS: Goods for Resale (P&L)
│       Cr Goods (BS Inventories) ← Stock reduces
│
├── POS Session Close (Cash Counted):
│       Dr Bank Account (BS)
│       Cr Cash on Hand (BS)
│       (Cash counted and deposited to bank)
│
└── POS Refund:
        Reversal of sale entry + re-instate inventory

BS Impact: Cash ↑ (sales), Inventories ↓, VAT Payable ↑
P&L Impact: Sales Revenue ↑, COGS ↑
```

---

10. Financial Statement Assembly Engine

10.1 How the BS and P&L Are Built from the COA

The financial statement engine does NOT store pre-calculated financial statement figures. Instead, it **queries the journal entry lines in real time** and assembles the statements on demand.

```
FINANCIAL STATEMENT ASSEMBLY PROCESS
│
├── User requests: "Balance Sheet as of 31 Dec 2025"
│
├── STEP 1 — Identify the date boundary
│       All journal entries with entry_date ≤ 31 Dec 2025 are included
│
├── STEP 2 — Group by COA Section
│       SELECT gl_account.section, SUM(debit - credit) as balance
│       FROM journal_entry_lines
│       JOIN journal_entries ON entry_date ≤ date
│       JOIN chart_of_accounts ON account type = BS
│       GROUP BY section, parent_group
│
├── STEP 3 — Apply Balance Type Rules
│       For DEBIT accounts (Assets, Expenses):
│           Net Balance = Total Debits − Total Credits
│       For CREDIT accounts (Liabilities, Income):
│           Net Balance = Total Credits − Total Debits
│
├── STEP 4 — Organize into Statement Structure
│       Group by hierarchy: Root → Parent Group → Account
│       Apply subtotals and totals
```

```
│
├── STEP 5 — Validate BS Equation
│       Total Assets = Total Liabilities + Total Equity
│       (Should always be exactly equal — if not, error in entries)
│
└── STEP 6 — Return formatted statement with drill-down capability
        Double-click any line → see underlying journal entries
        Double-click dimension value → see breakdown by item/warehouse/etc.
```

10.2 How the P&L Connects to the BS at Year-End

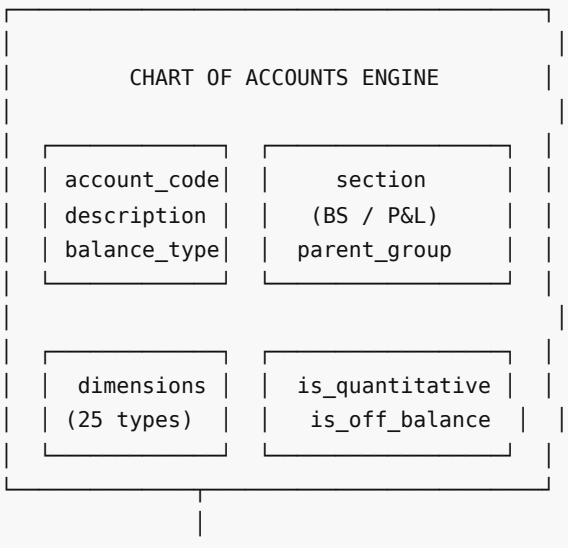
```
YEAR-END CLOSING PROCESS
│
├── During the year: Revenue and Expense accounts accumulate balances
│
├── At year-end: Closing journal entry is posted:
│       Dr All Revenue accounts (credit balances → zero them out)
│       Cr All Expense accounts (debit balances → zero them out)
│       Net:
│           If Revenue > Expenses → Cr Retained Earnings (BS Equity) ← PROFIT
│           If Expenses > Revenue → Dr Retained Earnings (BS Equity) ← LOSS
│
└── Result:
        P&L accounts reset to zero for the new year
        Profit/Loss permanently lives in BS Equity (Retained Earnings)

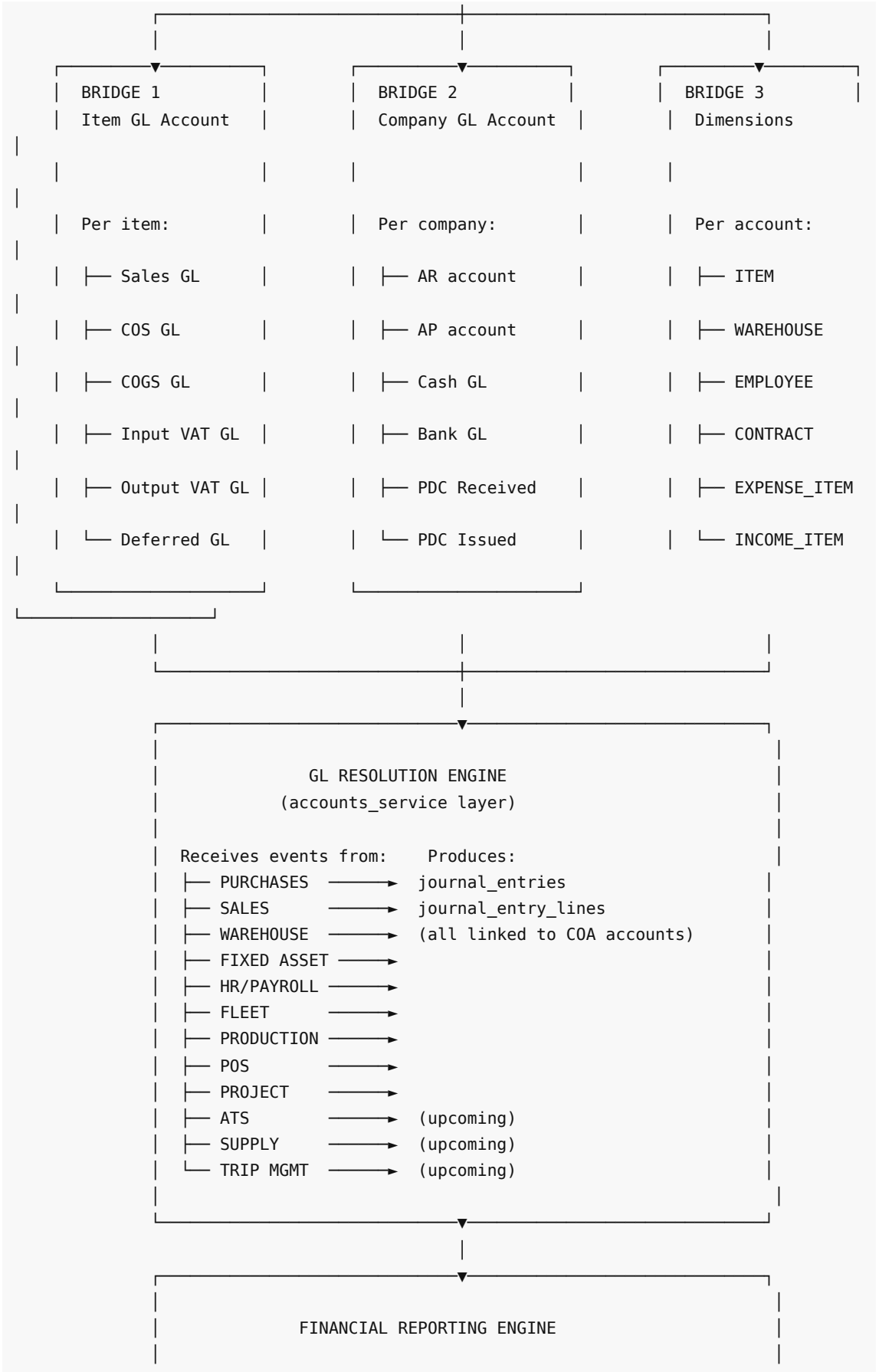
This is why the BS equation always holds:
    Assets = Liabilities + Equity
    And Equity grows with every profitable year
```

---

11. COA Integration Master Map

This is the complete visual showing how the COA Engine sits at the center and connects everything:




BRIDGE 1
Item GL Account
Per item:
├── Sales GL
├── COS GL
├── COGS GL
├── Input VAT GL
├── Output VAT GL
└── Deferred GL
BRIDGE 2
Company GL Account
Per company:
├── AR account
├── AP account
├── Cash GL
├── Bank GL
├── PDC Received
└── PDC Issued
BRIDGE 3
Dimensions
Per account:
├── ITEM
├── WAREHOUSE
├── EMPLOYEE
├── CONTRACT
├── EXPENSE_ITEM
└── INCOME_ITEM
GL RESOLUTION ENGINE
(accounts_service layer)
Receives events from:
Produces:
├── PURCHASES
├── SALES
├── WAREHOUSE
├── FIXED ASSET
├── HR/PAYROLL
├── FLEET
├── PRODUCTION
├── POS
├── PROJECT
├── ATS
├── SUPPLY
└── TRIP MGMT
journal_entries
journal_entry_lines
(all linked to COA accounts)
(upcoming)
(upcoming)
(upcoming)
FINANCIAL REPORTING ENGINE

```
│  ├── Trial Balance         ← All accounts, all periods       │
│  ├── General Ledger        ← All movements per account       │
│  ├── Balance Sheet         ← BS section grouping             │
│  ├── Income Statement      ← P&L section grouping            │
│  ├── Cash Flow Statement   ← CASH_FLOW_ITEM dimension        │
│  ├── Statement of Account  ← Per GL account / partner        │
│  └── VAT Reports           ← TAX_TYPE dimension              │
│                                                              │
└──────────────────────────────────────────────────────────────┘
```

Key Design Takeaways

Principle	How It Is Achieved
Zero coupling between modules and COA	Modules only fire events; GL Resolution Engine does all the accounting work
Any new module integrates in one step	Define the ItemGLAccount and/or CompanyGLAccount mapping; everything else is automatic
Financial statements are always live	No pre-aggregation; statements built on-demand from raw journal entries
Full drill-down capability	Every BS/P&L line traces back to individual journal entries through the COA section mapping
Analytical depth via dimensions	25 dimension types provide management accounting depth without additional accounts
Multi-tenant isolation	Each tenant has its own COA with its own account codes — no cross-tenant data is possible
IFRS compliance via sections	The `section` field on each account is the structural backbone of IFRS-compliant financial statements

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Focus: COA Engine Architecture — Dynamic Module Integration — BS & P&L Deep Dive
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